

DETECTING AND INVESTIGATING REMOTE DESKTOP PROTOCOL (RDP) BRUTE FORCE ATTACKS

Security Operation Center (SOC) Analysis Document

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Role Focus:

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Project Type:

Security Monitoring and Incident Detection

Security Stack:

Splunk (SIEM) | Windows Event Logs | pfsense

Objective:

To detect, analyze and generate alerts for RDP brute force authentication attempts using SIEM correlation rules and log analysis techniques

Date:

May 2026

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Incident Summary

A brute force attack is simulated from an attacker system running Kali Linux (192.168.2.5) targeting a windows server host (192.168.1.1) over Remote Desktop Protocol (RDP) on port 3389.

Incident Identification

- Incident Type: Brute Force Attack
- MITRE ATT&CK: T1110
- Target Service: Remote Desktop Protocol
- Destination Port: 3389
- Severity: Medium

Environment Details

Attacker System:

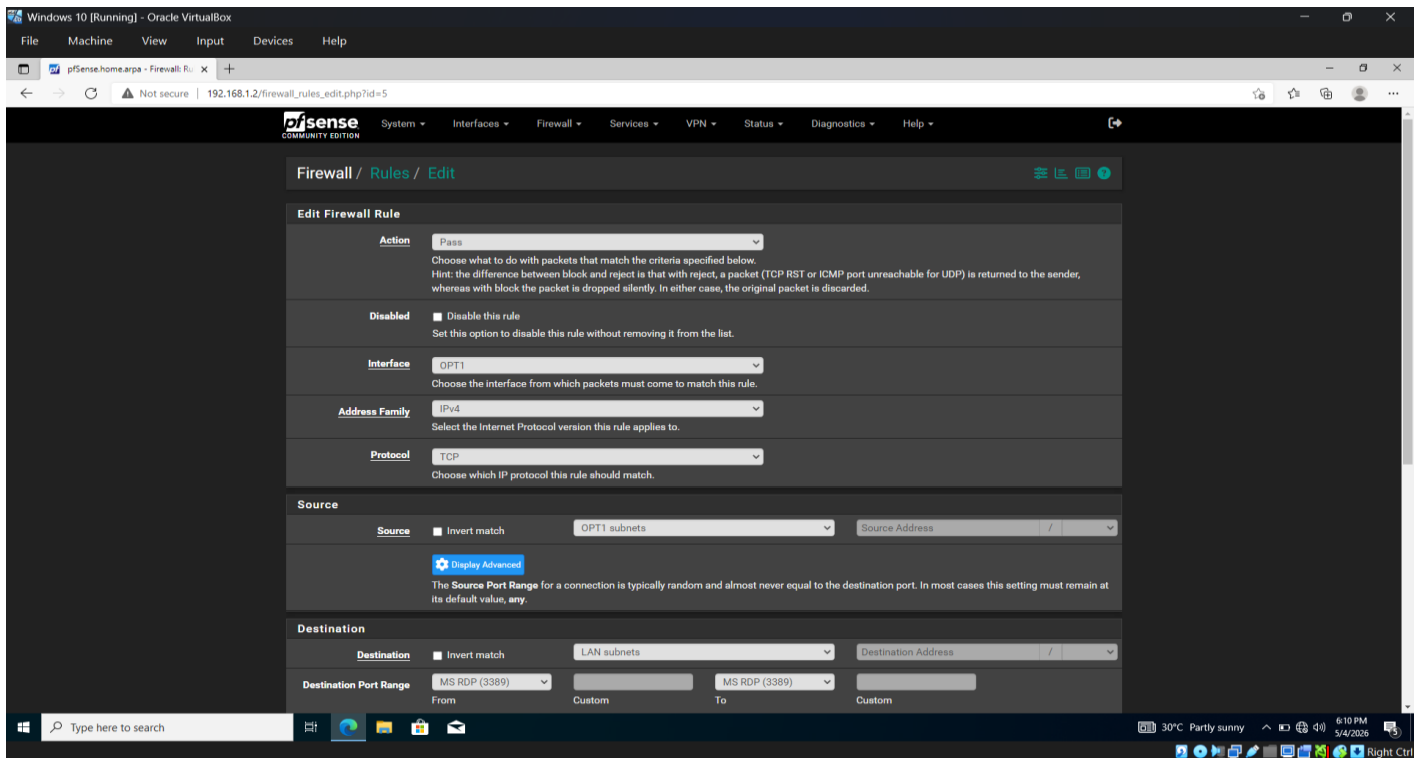
- Operating System: Kali Linux
- IP address: 192.168.2.5
- Network Zone: OPT1

Target System:

- Operating System: Windows Server
- IP address: 192.168.1.1
- RDP (port 3389) exposed
- Account Lockout Policy (5 attempts)
- Weak password

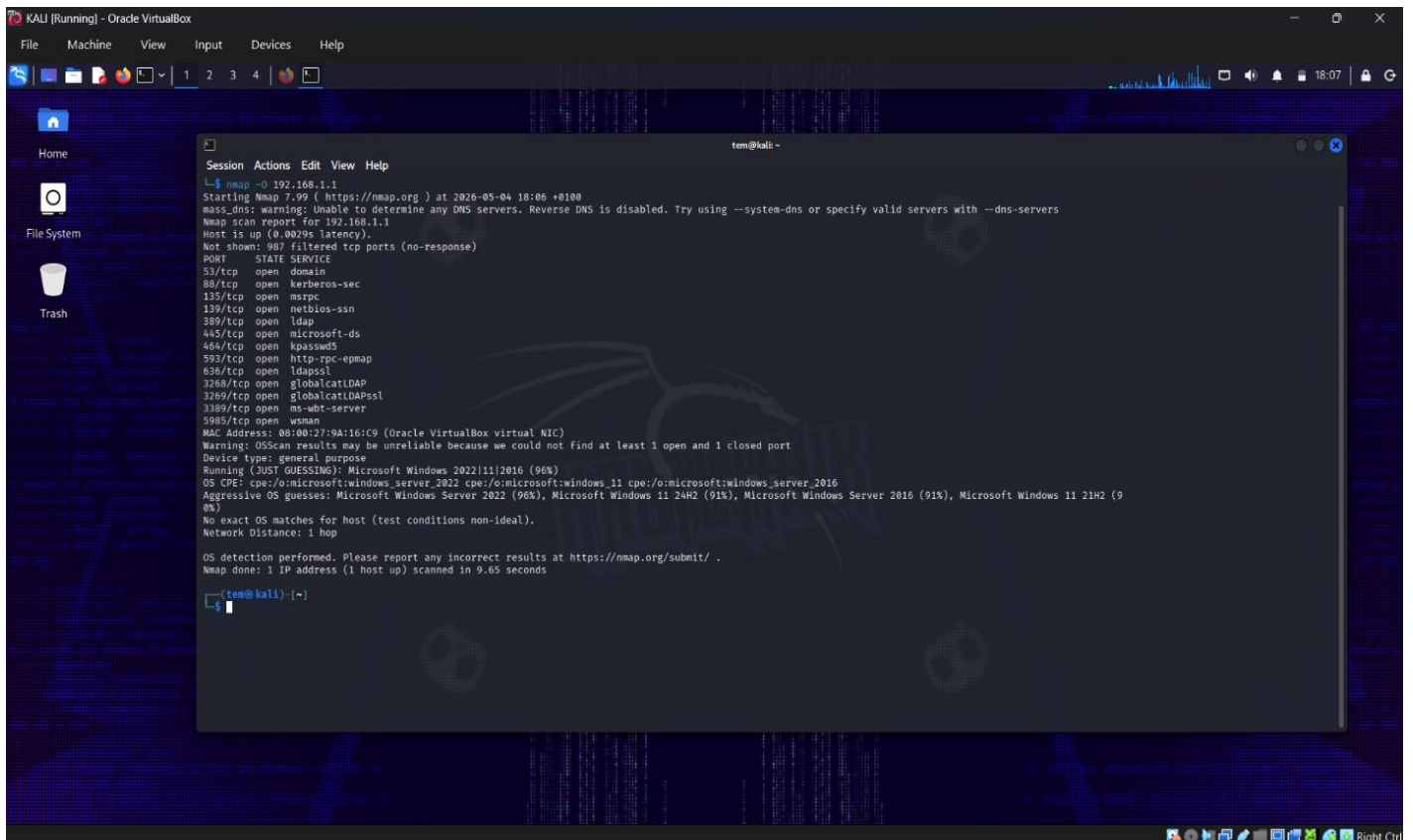
Firewall:

- Firewall rule configured to allow traffic
 - Action: Pass
 - Source Family: IPV4
 - Protocol: TCP
 - Source: OPT1 Subnet
 - Destination: LAN Subnet
 - Destination Port: MS RDP (3389)



Attack Simulation

- Network discovery is conducted using Nmap to identify open ports and exposed service.



- Using a tool called Hydra, the attacker initiates a brute force attack against the victim on port 3389

Detection Method

- Detection was performed using SIEM (Splunk) and firewall logs
- Alert was created to detect brute force attacks, attempts and also to detect lockouts
- High volume of authentication attempts
- Repeated login failures

The screenshot shows the Splunk Enterprise search interface. The search bar contains a complex query: `index=* sourcetype="WinEventLog:Security" (EventCode=4625 OR EventCode=4625 OR EventCode=4640) | eval user=coalesce(TargetUserName, Account_Name) | eval src=coalesce(IpAddress, Workstation_Name) | where isnotnull(user) | bin _time span=5m | stats count(eval(EventCode=4625)) as failed_attempts count(eval(EventCode=4640)) as successful_logins count(eval(EventCode=4640)) as lockouts min(_time) as first_seen max(_time) as last_seen by _time, user, src | eval duration = last_seen - first_seen | eval attack_pattern=case(failed_attempts>8 AND successful_logins>1, "Brute Force Attack", failed_attempts>8 AND successful_logins=0, "Brute Force Attempts", failed_attempts>5, "Suspicious Activity") | eval severity=case(attack_pattern="Brute Force Attack", "Critical", attack_pattern="Brute Force Attempts", "High", attack_pattern="Suspicious Activity", "Medium") | eval mitre_technique="Brute Force" | eval mitre_tactic="Credential Access" | where failed_attempts>3 | sort - failed_attempts`

The search results table shows two entries for failed login attempts from a Kali machine to a Windows server.

_time	user	src	failed_attempts	successful_logins	lockouts	first_seen	last_seen	attack_pattern	duration	mitre_tactic	mitre_technique	mitre_technique_id	severity
2020-05-04 16:25:00	-	kali	5	0	0	1777988300	1777988300	Suspicious Activity	0	Credential Access	Brute Force	T1110	Medium
2020-05-04 16:25:00	Administrator	kali	5	0	0	1777988300	1777988300	Suspicious Activity	0	Credential Access	Brute Force	T1110	Medium

- From the image, the source (Kali), User (Administrator which is windows server). There are 5 failed attempts from the same src hitting the same user, MITRE pattern (Suspicious Activity), severity is Medium.

KALI [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Search | Splunk 10.2.2

Not Secure http://192.168.1.10:8000/en-US/app/search/search?display.page.search.mode=fast&dispatch.sample_ratio=1&q=search index%3D* EventCode%3D4625 | eval src%3Dcoalesce(ipAd

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< Hide Fields All Fields Format Show: 20 Per Page View: List

sourcetype 1

INTERESTING FIELDS

EventCode 1

index 1

linecount 1

Message 1

Source_Network_Address 1

splunk_server 1

src 1

+ Extract New Fields

Time	Event
	SourceName=Microsoft Windows security auditing. Type=Information RecordNumber=125985 Keywords=Audit Failure TaskCategory=Process Creation OpCode=Info Message=An account failed to log on. Subject: Security ID: S-1-0-0 Account Name: - Account Domain: - Logon ID: 0x0 Logon Type: 3 Account For Which Logon Failed: Security ID: S-1-0-0 Account Name: Administrator Account Domain: - Failure Information: Failure Reason: Unknown user name or bad password. Status: 0xC000006D Sub Status: 0xC000006A Process Information: Caller Process ID: 0x0 Caller Process Name: - Network Information: Workstation Name: kali Source Network Address: 192.168.2.5 Source Port: 0 Detailed Authentication Information: Logon Process: NtLmSsp Authentication Package: NTLM

Right Ctrl

KALI [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Search | Splunk 10.2.2

Not Secure http://192.168.1.10:8000/en-US/app/search/search?display.page.search.mode=fast&dispatch.sample_ratio=1&q=search index%3D* EventCode%3D4625 | eval src%3Dcoalesce(ipAd

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< Hide Fields All Fields Format Show: 20 Per Page View: List

host 1

source 1

sourcetype 1

INTERESTING FIELDS

EventCode 1

index 1

linecount 1

Message 1

Source_Network_Address 1

splunk_server 1

src 1

+ Extract New Fields

Time	Event
	ComputerName=Server.cis.net LogName=Security EventCode=4625 EventType=0 ComputerName=Server.cis.net host = Server source = WinEventLog\Security sourcetype = WinEventLog\Security
5/4/26 4:26:37.040 PM	LogName=Security EventCode=4625 EventType=0 ComputerName=Server.cis.net host = Server source = WinEventLog\Security sourcetype = WinEventLog\Security
5/4/26 4:26:32.651 PM	LogName=Security EventCode=4625 EventType=0 ComputerName=Server.cis.net host = Server source = WinEventLog\Security sourcetype = WinEventLog\Security
5/4/26 4:26:25.434 PM	LogName=Security EventCode=4625 EventType=0 ComputerName=Server.cis.net host = Server source = WinEventLog\Security sourcetype = WinEventLog\Security
5/4/26 4:26:17.955 PM	LogName=Security EventCode=4625 EventType=0 ComputerName=Server.cis.net host = Server source = WinEventLog\Security sourcetype = WinEventLog\Security
5/4/26 7:53:52.679 AM	LogName=Security

Right Ctrl

Status	Time	Interface	Source IP	Destination IP	Protocol
X	Apr 19 08:27:12	OPT1	192.168.1.1	224.0.0.22	IGMP
X	Apr 19 08:27:13	OPT1	[fe80::b8c9bd7c:3b05:ddc5]:546	[ff02::1:2]:547	UDP
X	Apr 19 08:27:13	LAN	[fe80::b8c9bd7c:3b05:ddc5]:546	[ff02::1:2]:547	UDP
✓	Apr 19 08:27:14	OPT1	192.168.2.5:34278	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:14	OPT1	192.168.2.5:34284	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:15	OPT1	192.168.2.5:34286	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:17	OPT1	192.168.2.5:40490	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:17	OPT1	192.168.2.5:40502	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:18	OPT1	192.168.2.5:40508	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:19	OPT1	192.168.2.5:40516	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:21	OPT1	192.168.2.5:40528	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:22	OPT1	192.168.2.5:40532	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:24	OPT1	192.168.2.5:40534	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:24	OPT1	192.168.2.5:40538	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:26	OPT1	192.168.2.5:40548	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:28	OPT1	192.168.2.5:41772	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:29	OPT1	192.168.2.5:41774	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:29	OPT1	192.168.2.5:41784	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:30	OPT1	192.168.2.5:41788	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:32	OPT1	192.168.2.5:41804	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:32	OPT1	192.168.2.5:41806	192.168.1.1:3389	TCP-S
✓	Apr 19 08:27:34	OPT1	192.168.2.5:41822	192.168.1.1:3389	TCP-S
X	Apr 19 08:27:45	OPT1	[fe80::b8c9bd7c:3b05:ddc5]:546	[ff02::1:2]:547	UDP
X	Apr 19 08:27:45	LAN	[fe80::b8c9bd7c:3b05:ddc5]:546	[ff02::1:2]:547	UDP
X	Apr 19 08:28:50	LAN	192.168.1.3:60604	239.255.255.250:1900	UDP

Analysis

The activity matches brute behavior due to:

- Repeated authentication failures
- Consistence source IP (192.168.2.5) to same user (192.168.1.1)
- Targeting of RDP
 - Risk increases significantly if:
 - A successful login occurs

IMPACT ASSESSMENT

Confidential impact

- The attacker successfully gained access via RDP which lead to:
 - Unauthorized access to sensitive data (in an organization for example, customer's records, credentials, internal documents)
 - Potential data exfiltration leading to regulatory violations
 - BUSINESS CONSEQUENCES:
 - Loss customers trust
 - Legal penalties

Integrity Impacts

With valid access, attacker could:

- Modify system configurations
- Inject malicious scripts or backdoors
 - BUSINESS CONSEQUENCES:
 - Corrupted business data
 - Loss of trust in system accuracy

Availability Impact

- Customers may not have access to their data
 - BUSINESS CONSEQUENCES
 - Operational downtime
 - Disruption of business services
 - Revenue loss due to service unavailability

Containment, Eradication and Recovery

- Block source IP at firewall
- Restrict RDP exposure
- Disable or lock targeted accounts
- Reset credentials
- Validate system integrity
- Enforce strong password policies
- Disable direct RDP exposure to untrusted networks

Lesson Learned

- Exposed RDP significantly increases attack surface
- Always enable strong passwords (characters, numbers and values not less than 10)
- Detection must focus on failed attempts not only success.